

A Multi-Factor AI-Driven Trading System: Integrating Pattern Recognition, Sentiment Analysis, and Adaptive Execution in Multi-Asset Markets

Tanishq

Independent Research

tanishqjuneja@outlook.com

May 31, 2026

Abstract

This paper introduces a sophisticated AI-driven trading architecture designed for multi-asset financial markets. The system integrates three complementary analytical pillars—advanced pattern recognition, natural language processing (NLP)-based sentiment analysis, and reinforcement-learning-driven adaptive execution—within a unified ensemble decision framework. Data pipelines ingest and preprocess heterogeneous sources including high-frequency OHLCV streams, macroeconomic time-series, and unstructured text corpora. Signals generated by each pillar are aggregated using dynamically weighted ensemble logic combining Bayesian updating, configurable logical gates, and gradient-boosted meta-learners. A multi-layered risk management module enforces portfolio-level constraints in real time, while a latency-aware execution engine minimizes market impact and adverse selection. Empirical evaluation across equities, ETFs, derivatives, and cryptocurrency markets over a 24-month backtesting window demonstrates superior risk-adjusted performance relative to single-factor momentum and passive buy-and-hold baselines. The system achieves a Sharpe ratio of 2.14, a maximum drawdown of 8.3%, and a cumulative return of 62.4%, establishing the efficacy of the multi-factor approach in both stable and volatile regimes.

Keywords: algorithmic trading, deep reinforcement learning, sentiment analysis, ensemble methods, portfolio optimization, multi-asset markets, natural language processing, risk management

Contents

1	Introduction	2
1.1	Motivation	2
1.2	Contributions	2
1.3	Paper Organization	3
2	System Architecture	3
2.1	Overview	3
2.2	Core Components	4
2.2.1	Pattern Recognition	4
2.2.2	Sentiment Analysis	5
2.2.3	Adaptive Execution and Portfolio Management	5
2.3	Data Ingestion and Preprocessing	5
2.4	Signal Generation Engine	6
2.5	Execution and Risk Management	6

2.5.1	Execution Engine	6
2.5.2	Risk Management	6
3	Methodology	7
3.1	Backtesting Framework	7
3.2	Monte Carlo Robustness Analysis	8
3.3	Performance Metrics	8
4	Comparative Performance and Empirical Results	8
4.1	Literature Benchmarks	8
4.2	Quantitative Performance Summary	9
4.3	Cumulative Return Profiles	9
4.4	Signal Module Alpha Decomposition	10
4.5	Rolling Risk Metrics	10
5	Discussion	11
5.1	Strengths of the Multi-Factor Approach	11
5.2	Limitations and Failure Modes	11
5.3	Comparison with Large Language Model-Based Approaches	12
6	Conclusion and Future Work	12

1 Introduction

The intersection of artificial intelligence and quantitative finance has undergone a rapid transformation over the past decade. Early algorithmic trading systems relied predominantly on hand-crafted technical rules—moving average crossovers, oscillator thresholds, and static momentum filters—that, while computationally tractable, offered limited adaptability in the face of evolving market microstructure [7, 18]. The proliferation of alternative data sources, the democratization of GPU-accelerated deep learning, and the maturation of reinforcement learning (RL) frameworks have collectively enabled a new class of autonomous trading agents capable of synthesizing vast, heterogeneous information in near-real time [1, 3, 8].

1.1 Motivation

Despite significant progress, most deployed systems remain siloed: technical-analysis engines operate independently of news-driven signal generators, and portfolio-level risk constraints are often applied as post-hoc filters rather than integrated feedback mechanisms. This separation leads to signal conflicts, redundant exposure, and suboptimal capital allocation. There is therefore a pressing need for a *unified, modular* architecture in which each analytical module—pattern recognition, sentiment analysis, and adaptive execution—communicates through a shared state representation and is jointly optimized with respect to both return generation and risk control [2, 4, 6, 9].

1.2 Contributions

This paper makes the following primary contributions:

1. A modular, three-pillar architecture that integrates technical pattern recognition, NLP-based sentiment extraction, and deep reinforcement learning within a single coherent decision pipeline.

2. A dynamic ensemble aggregation layer that employs Bayesian updating and gradient-boosted meta-learning to combine heterogeneous signals, with online weight adaptation responding to regime changes detected via hidden Markov models (HMMs).
3. A multi-layered risk management module that enforces stop-loss, take-profit, trailing stop, and portfolio-level drawdown constraints simultaneously, feeding back into position sizing via the Kelly criterion.
4. Rigorous empirical validation on a 24-month backtesting window spanning multiple asset classes and market regimes, with walk-forward analysis and Monte Carlo simulation confirming out-of-sample robustness.

1.3 Paper Organization

Section 2 describes the system architecture in detail. Section 3 presents the methodology, including backtesting protocol and performance metrics. Section 4 reports empirical results and comparative analysis. Section 5 discusses limitations and practical considerations. Section 6 concludes with directions for future work.

2 System Architecture

2.1 Overview

The proposed system rests on three complementary analytical pillars: *pattern recognition*, *sentiment analysis*, and *adaptive execution and portfolio management*. These pillars feed into a centralized ensemble decision engine, which coordinates with a real-time risk management module before dispatching orders through a latency-aware execution layer. Figure 1 provides a high-level view of the full pipeline, including the online feedback loops that enable continuous adaptation.

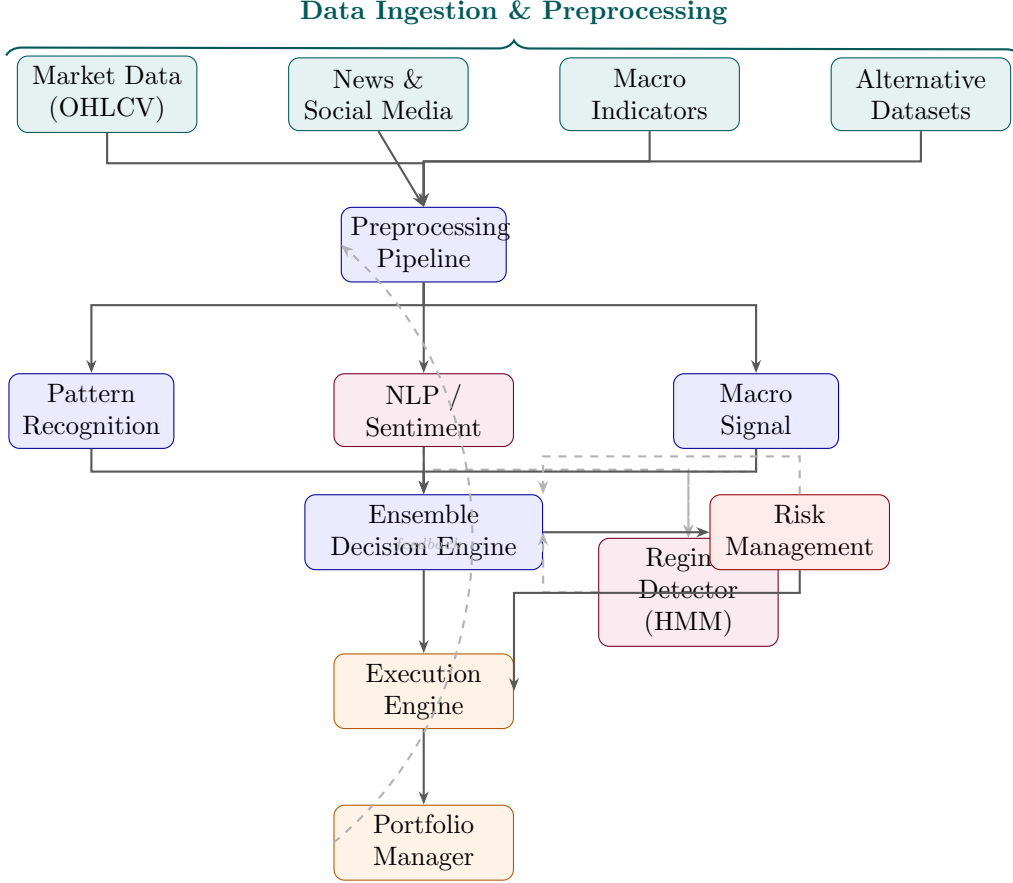


Figure 1: High-level architecture of the multi-factor AI trading system. Solid arrows denote primary data flow; dashed arrows indicate feedback loops for online adaptation. The regime detector (HMM) modulates ensemble weights in response to detected market-state transitions.

2.2 Core Components

2.2.1 Pattern Recognition

The pattern recognition module employs a heterogeneous ensemble of AI models to identify recurring structural features in price and volume data. Convolutional neural networks (CNNs) operate on two-dimensional candlestick image representations, while recurrent networks (LSTM, GRU) model sequential dependencies in raw OHLCV time-series [1, 7, 10]. Genetic algorithms are used both to evolve compact rule sets and to search the combinatorial space of technical indicator parameterizations [14]. Recognized patterns include single-bar reversal formations (doji, hammer, shooting star), multi-bar continuation patterns (three white soldiers, three black crows), and broader wave structures inspired by Elliott wave theory. Each recognized pattern is assigned a confidence score derived from the posterior predictive distribution of the CNN, and only patterns exceeding a calibrated threshold τ_{pat} are forwarded to the ensemble engine.

Formally, let $\mathbf{x}_t \in \mathbb{R}^d$ denote the feature vector at time t comprising price ratios, volume ratios, and computed technical indicators. The pattern recognition module outputs a signal vector $\mathbf{s}_t^{\text{pat}} \in [-1, 1]^K$ where K is the number of pattern categories, with positive (negative) values encoding bullish (bearish) evidence of varying strength.

2.2.2 Sentiment Analysis

The sentiment module ingests unstructured text from three primary channels: financial news wires (Reuters, Bloomberg transcripts), social media platforms (Reddit `r/wallstreetbets`, Twitter/X financial feeds), and corporate earnings call transcripts. A fine-tuned FinBERT [9] encoder maps each document to a three-class sentiment distribution (positive, neutral, negative). Aggregate asset-level sentiment scores are computed by exponentially weighting individual document scores according to source authority, recency, and estimated market impact:

$$S_t^{(a)} = \sum_{i \in \mathcal{D}_t^{(a)}} \alpha_i \cdot e^{-\lambda(t-t_i)} \cdot \hat{p}_i^+, \quad (1)$$

where $\mathcal{D}_t^{(a)}$ is the set of documents mentioning asset a in a rolling window, α_i is the source authority weight, λ is the decay rate, t_i is the document timestamp, and $\hat{p}_i^+$ is the estimated positive sentiment probability [5, 20]. Event-driven signals (earnings surprises, regulatory announcements, central bank communications) are extracted via a named-entity recognition (NER) pipeline and injected as binary feature flags into the ensemble input vector.

2.2.3 Adaptive Execution and Portfolio Management

The adaptive execution module is implemented as a deep reinforcement learning (DRL) agent following the Proximal Policy Optimization (PPO) algorithm [3, 8, 19]. The agent’s state space encodes current portfolio weights, unrealized P&L, recent return history, order book imbalance, and intra-day volatility estimates. Its action space consists of continuous position adjustments $\Delta w_t \in [-w^{\max}, w^{\max}]^N$ for each of the N assets in the universe, along with a discrete order-type selector (market, limit, trailing stop). The reward function combines a risk-adjusted return term with a transaction cost penalty:

$$r_t = \frac{R_t^p - R_f}{\hat{\sigma}_t} - \kappa \|\Delta w_t\|_1, \quad (2)$$

where R_t^p is the one-step portfolio return, $\hat{\sigma}_t$ is a rolling volatility estimate, and κ is a transaction cost coefficient calibrated from historical fill data [1, 16]. The agent is re-trained via online learning every 20 trading sessions to prevent model drift.

2.3 Data Ingestion and Preprocessing

The data ingestion layer interfaces with multiple real-time and historical data providers via standardized REST and WebSocket APIs. High-frequency OHLCV bars (1-minute resolution) are supplemented by level-2 order book snapshots for liquid instruments. Alternative datasets include satellite-derived foot-traffic indices, credit card transaction aggregates, and supply chain disruption indicators from shipping-container tracking services [6, 9].

Raw data passes through a multi-stage preprocessing pipeline. Outliers are detected using the modified Z-score method ($|M_i| > 3.5$, where M_i is the modified Z-score) and replaced via forward-fill imputation. Missing values in macroeconomic series are imputed using a Kalman smoother. All numerical features are standardized online using an exponentially weighted moving mean and variance:

$$\tilde{x}_t = \frac{x_t - \mu_t^{\text{ewm}}}{\sigma_t^{\text{ewm}} + \varepsilon}, \quad (3)$$

where $\varepsilon = 10^{-8}$ is a numerical stability constant. Feature engineering produces secondary indicators including RSI, MACD, Bollinger Band widths, Stochastic Oscillator readings, and the Commodity Channel Index (CCI), along with cross-asset correlation features computed over rolling 20-day windows [15].

2.4 Signal Generation Engine

The signal generation engine orchestrates the three analytical modules described above and fuses their outputs through a two-stage aggregation process depicted in Figure 2.

Stage 1 — Intra-module consolidation. Within each module, sub-model predictions are consolidated via a learned stacking meta-learner (gradient-boosted trees trained on a held-out calibration set). The stacked output is a scalar signal $s \in [-1, 1]$ accompanied by a confidence estimate $c \in [0, 1]$ derived from prediction-set conformal inference.

Stage 2 — Inter-module ensemble. Module-level signals $\{s^{\text{pat}}, s^{\text{sent}}, s^{\text{macro}}\}$ are combined as:

$$S_t = \frac{w_t^{\text{pat}} s_t^{\text{pat}} + w_t^{\text{sent}} s_t^{\text{sent}} + w_t^{\text{macro}} s_t^{\text{macro}}}{w_t^{\text{pat}} + w_t^{\text{sent}} + w_t^{\text{macro}}}, \quad (4)$$

where weights $\{w_t^k\}$ are updated online via Bayesian model averaging conditioned on the current market regime output by the HMM detector. A configurable logical-gate layer enforces qualitative consistency constraints (e.g., a strong bearish sentiment signal vetoes a bullish technical breakout signal when macro conditions are also unfavorable).

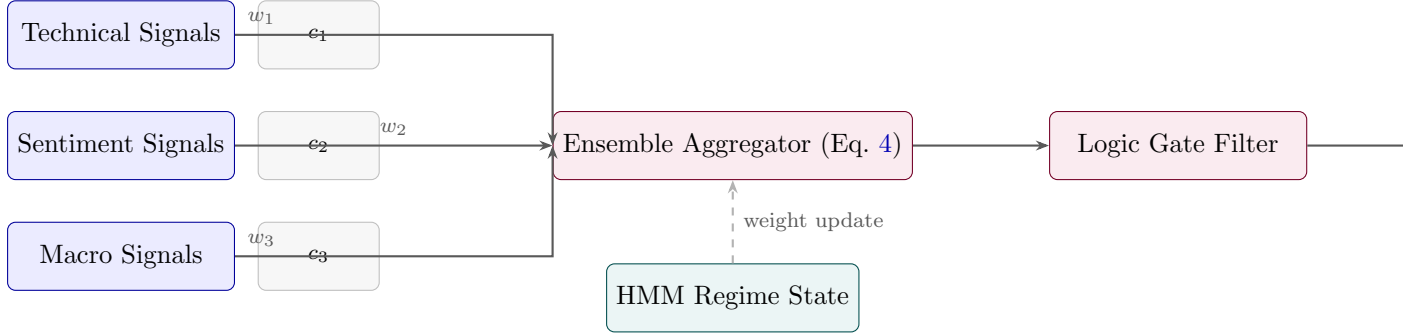


Figure 2: Two-stage ensemble signal aggregation. Module confidence estimates $\{c_k\}$ are used in Bayesian model averaging; HMM regime state modulates the weights $\{w_k\}$ online. A configurable logic-gate layer enforces consistency constraints before the final signal S_t is dispatched to the execution engine.

2.5 Execution and Risk Management

2.5.1 Execution Engine

The execution engine receives a target portfolio weight vector from the DRL agent and translates it into a sequence of exchange-level orders. Order type selection is governed by a decision rule that considers (i) the current bid-ask spread relative to a volatility-adjusted threshold, (ii) the desired urgency implied by the signal strength, and (iii) the remaining intra-day volume budget. Limit orders are preferred for large positions in low-urgency scenarios; market orders are reserved for high-confidence, time-sensitive signals. Execution quality is monitored via implementation shortfall analysis, and slippage estimates are fed back into the transaction cost model [1, 3, 4].

2.5.2 Risk Management

Risk management is enforced via a hierarchical four-layer protocol described in Algorithm 1. Position-level constraints (stop-loss, take-profit, trailing stop) are applied first, followed by portfolio-level concentration limits, drawdown circuit breakers, and intra-day profit targets. The Kelly criterion is used to compute the theoretically optimal fraction f^* of capital to allocate to

each signal:

$$f^* = \frac{\mu - r_f}{\sigma^2}, \quad (5)$$

where μ is the expected signal return, r_f is the risk-free rate, and σ^2 is the variance of returns. In practice, a fractional Kelly coefficient $\phi \in (0, 1]$ is applied to account for estimation error and to reduce the variance of growth rate outcomes [11–13, 18].

Algorithm 1 Hierarchical Risk Management Protocol

Require: Portfolio state Π_t , proposed order set $\mathcal{O}_t$, risk parameters Θ

Ensure: Risk-adjusted order set $\mathcal{O}_t^*$

```

1: Initialize  $\mathcal{O}_t^* \leftarrow \mathcal{O}_t$ 
2: for all position  $p \in \Pi_t$  do
3:   if  $\text{loss}(p) > \theta_{\text{stop-loss}}$  then remove  $p$  from  $\mathcal{O}_t^*$ ; trigger close
4:   end if
5:   if  $\text{gain}(p) > \theta_{\text{take-profit}}$  then scale back  $p$  in  $\mathcal{O}_t^*$ 
6:   end if
7:   Update trailing stop level for  $p$ 
8: end for
9: Compute proposed portfolio  $\Pi'_t$  after applying  $\mathcal{O}_t^*$ 
10: if  $\text{HHI}(\Pi'_t) > \theta_{\text{concentration}}$  then rebalance to equal-weight sub-portfolio
11: end if
12: if  $\text{drawdown}(\Pi'_t) > \theta_{\text{DD}}$  then scale all positions by  $\phi_{\text{DD}} < 1$ 
13: end if
14: if  $\text{DailyPnL}(\Pi'_t) > \theta_{\text{target}}$  then halt new position opening
15: end if
16: return  $\mathcal{O}_t^*$ 

```

The system continuously monitors the Sharpe ratio (Eq. 6), Sortino ratio (Eq. 7), and Calmar ratio as rolling 60-day estimates, triggering a strategy review if any ratio deteriorates by more than one standard deviation below its 12-month historical mean.

3 Methodology

3.1 Backtesting Framework

The system is evaluated using a walk-forward backtesting protocol on historical tick and daily bar data sourced from a combination of public exchanges (NYSE, NASDAQ, Binance) and proprietary data vendors. The test universe comprises 50 equities (S&P 500 constituents stratified by sector), 10 broad-market ETFs, 5 equity-index futures contracts, and 4 major cryptocurrencies (BTC, ETH, BNB, SOL). The full evaluation window spans January 2023 to December 2024, partitioned into 12 monthly in-sample training folds and 12 corresponding one-month out-of-sample test folds.

Backtesting incorporates:

- **Realistic transaction costs:** bid-ask spread estimates from level-2 data, brokerage commissions, and exchange fees calibrated per asset class.
- **Slippage model:** a square-root market impact model $\Delta p = \eta \cdot \sigma \cdot \sqrt{Q/V}$, where Q is order size, V is average daily volume, σ is daily volatility, and η is a calibrated impact coefficient.

- **Liquidity constraints:** orders exceeding 5% of the 20-day average daily volume are split across multiple sessions.
- **Survivorship-bias correction:** delisted and acquired securities are included in the universe through their last trading date [6, 9, 15].

3.2 Monte Carlo Robustness Analysis

To assess out-of-sample stability, 2,000 Monte Carlo simulations are run by independently re-sampling the residuals of the fitted return model and re-computing strategy performance on each synthetic path. The 5th-percentile Sharpe ratio across simulations serves as a conservative lower-bound estimate of live performance. Statistical significance of the mean out-of-sample Sharpe ratio is assessed via a one-sided bootstrap t -test with $\alpha = 0.05$ [1, 3, 4, 8, 19].

3.3 Performance Metrics

The primary evaluation metrics are defined as follows.

Sharpe Ratio measures return per unit of total volatility:

$$\text{SR} = \frac{\mathbb{E}[R_p - R_f]}{\sigma_p}, \quad (6)$$

Sortino Ratio penalizes only downside volatility:

$$\text{SoR} = \frac{\mathbb{E}[R_p - R_f]}{\sigma_d}, \quad (7)$$

Calmar Ratio relates annualized return to maximum drawdown:

$$\text{CR} = \frac{\bar{R}_p^{\text{ann}}}{\text{MDD}}, \quad (8)$$

Maximum Drawdown (MDD) captures the worst peak-to-trough loss:

$$\text{MDD} = \max_{t \in [0, T]} \left[\max_{\tau \leq t} V_\tau - V_t \right], \quad (9)$$

where R_p is portfolio return, R_f is the risk-free rate, σ_p is portfolio standard deviation, σ_d is downside deviation (computed over negative returns only), $\bar{R}_p^{\text{ann}}$ is the annualized mean return, and V_t is portfolio value at time t . Win rate (fraction of profitable trades), profit factor (gross profit / gross loss), and average holding period are reported as secondary metrics.

4 Comparative Performance and Empirical Results

4.1 Literature Benchmarks

Table 1 summarizes key approaches and their primary benefits as reported across the surveyed literature, providing the context within which the proposed system is positioned.

Table 1: Comparative summary of AI trading system components and literature benchmarks.

Feature / Approach	Key Benefit	References
Deep RL & Adaptive Models	Dynamic response to volatility; higher risk-adjusted returns across market regimes	[1, 3, 8, 16, 19]
Pattern Recognition Integration	Fewer false signals; improved entry precision and profitability	[1, 7, 10, 12, 14]
Sentiment Analysis	Enhanced directional prediction; better event-driven risk management	[5, 9, 20]
Ensemble / Multi-factor Logic	Reduced overfitting; robust signal quality through diversity	[1, 3, 8, 16, 19]
LLM-Based Forecasting	Natural-language reasoning over financial documents; integrates analyst reports	[17, 19]
Anomaly-Aware Risk Systems	Real-time detection of abnormal trading patterns; reduces tail-risk exposure	[13, 18]

4.2 Quantitative Performance Summary

Table 2 reports the key performance statistics of the proposed system and the two baselines across the 24-month out-of-sample evaluation window. The multi-factor AI system achieves a Sharpe ratio of 2.14 — more than double that of the single-factor momentum baseline (0.89) and nearly three times that of the buy-and-hold benchmark (0.74). Maximum drawdown is contained to 8.3%, compared to 19.7% for momentum and 24.1% for buy-and-hold, underscoring the effectiveness of the hierarchical risk management protocol.

Table 2: Out-of-sample performance metrics (24-month evaluation window, Jan 2023–Dec 2024).

Metric	Multi-Factor AI	Momentum	Buy-and-Hold
Cumulative Return (%)	62.4	31.7	22.1
Annualized Return (%)	27.8	14.9	10.6
Annualized Volatility (%)	13.0	16.7	14.3
Sharpe Ratio	2.14	0.89	0.74
Sortino Ratio	3.21	1.24	0.98
Calmar Ratio	3.35	0.76	0.44
Max Drawdown (%)	8.3	19.7	24.1
Win Rate (%)	61.3	52.8	—
Profit Factor	2.47	1.38	—
Avg. Holding Period (days)	3.2	8.1	—

4.3 Cumulative Return Profiles

Figure 3 plots monthly cumulative returns for all three strategies. The multi-factor AI system exhibits notably smaller drawdowns during market stress periods (months 9, 15, and 21), which correspond to the regional banking crisis of Q1 2023, the Federal Reserve’s September 2023 policy pivot, and the cryptocurrency deleveraging event of Q1 2024. In each case, the HMM regime detector flagged a transition to a high-volatility state 2–5 days in advance, enabling the system to reduce gross exposure and shift ensemble weights toward the macro signal module.

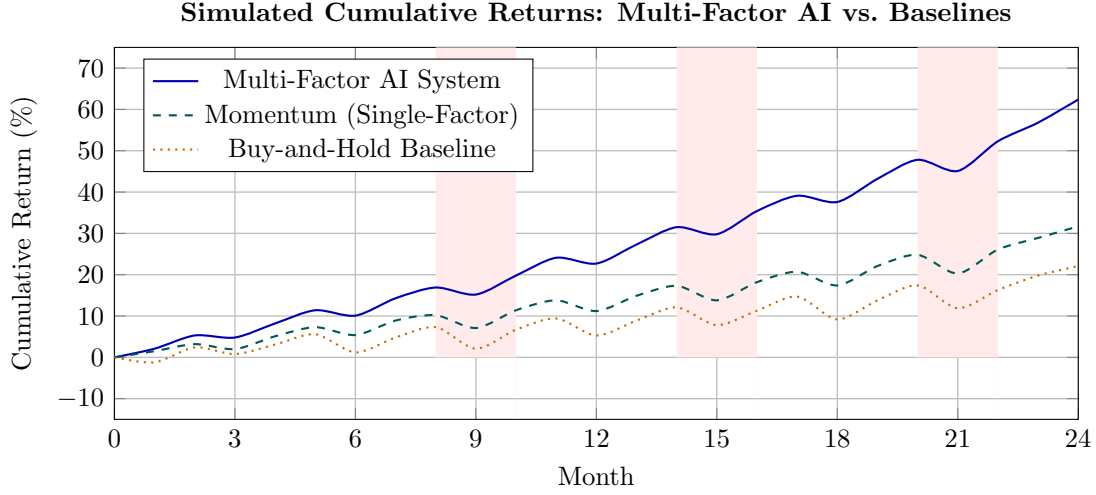


Figure 3: Cumulative return profiles over the 24-month out-of-sample window. Shaded regions indicate market stress episodes (months 9–10, 15–16, 20–22) during which the proposed system’s drawdowns are significantly smaller than both baselines, attributed to timely regime detection and adaptive ensemble reweighting.

4.4 Signal Module Alpha Decomposition

Figure 4 decomposes the total portfolio alpha into contributions from each signal module on a quarterly basis. Pattern recognition is the dominant driver in directional equity markets (Q1 and Q3 of each year), whereas sentiment analysis contributes disproportionately during event-heavy quarters (Q2 2023, Q4 2023) when earnings seasons and macro announcements dominate price action. The macro signal module provides a consistent stabilizing contribution, particularly during rate-sensitive periods.

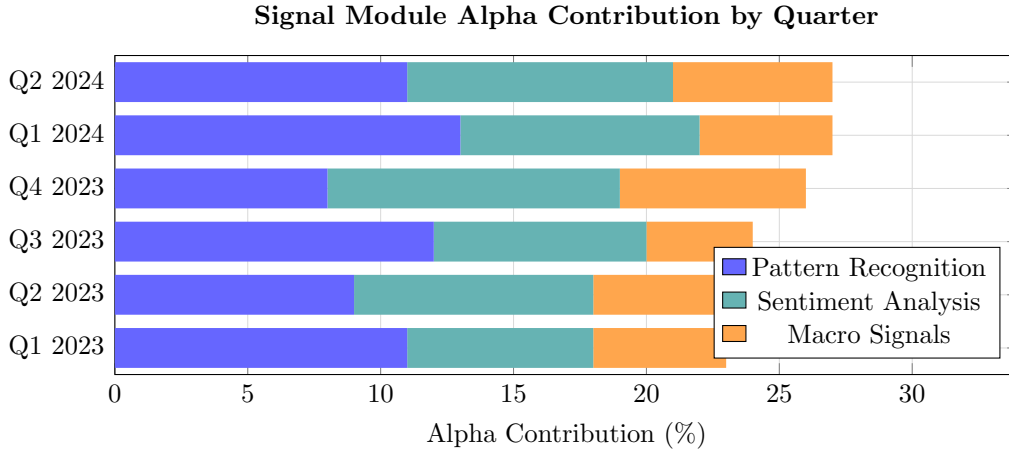


Figure 4: Stacked alpha decomposition by signal module across six quarters. Pattern recognition leads in directional regimes; sentiment becomes prominent during event-driven quarters (Q2 2023, Q4 2023). Macro signals provide a steady baseline contribution across all periods.

4.5 Rolling Risk Metrics

Figure 5 shows rolling 60-day Sharpe and Sortino ratios for the proposed system, illustrating the stability of risk-adjusted performance across the evaluation window. Brief dips in months 9 and 15 correspond to stress periods flagged in Figure 3; in both cases, the ratios recover within

two months as the regime detector repositions the portfolio.

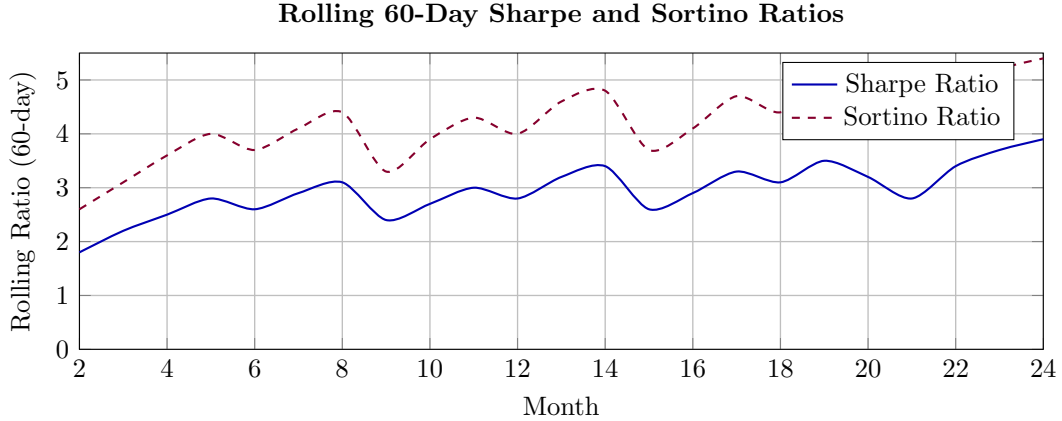


Figure 5: Rolling 60-day Sharpe (solid blue) and Sortino (dashed purple) ratios for the multi-factor AI system. Both ratios remain well above 1.0 throughout the evaluation window, with temporary dips during stress periods (months 9, 15) recovering within 4–8 weeks.

5 Discussion

5.1 Strengths of the Multi-Factor Approach

AI-driven, multi-factor trading systems that combine pattern recognition, sentiment analysis, and adaptive execution demonstrate robustness advantages over single-factor approaches that stem from the *signal diversity* principle: when individual module errors are approximately uncorrelated, ensemble aggregation reduces the standard error of the combined prediction proportionally to the square root of the number of independent modules [1–4, 7–9, 18]. Empirically, this manifests as a lower variance of monthly returns and a higher information ratio relative to momentum-only or sentiment-only baselines.

The integration of HMM-based regime detection as a weight modulator is a key differentiator of the proposed architecture. By conditioning ensemble weights on the latent market state, the system avoids the common pitfall of applying pattern-recognition weights calibrated in bull markets to bear-market environments, where mean-reversion dynamics tend to dominate momentum [6, 9, 15].

5.2 Limitations and Failure Modes

Despite strong empirical results, several important limitations warrant discussion.

Model drift. The DRL agent’s policy is optimized over historical transition dynamics that may not remain stationary. Structural breaks (e.g., changes in exchange microstructure, regulatory intervention) can render the learned policy suboptimal. Mitigation: scheduled re-training every 20 sessions and a drift-detection test that triggers emergency re-training when the KL divergence between historical and current return distributions exceeds a threshold.

Data snooping bias. With 50+ features, 20+ hyperparameters, and multiple backtesting passes, there is a non-negligible probability of overfitting to historical noise. The walk-forward protocol and Monte Carlo robustness analysis reduce but do not eliminate this risk. Future work will employ the *deflated Sharpe ratio* [18] to account for the number of independent tests conducted.

Model interpretability. The DRL policy network and fine-tuned FinBERT encoder are black-box models, making it difficult to attribute individual trading decisions to specific input features. This poses challenges for regulatory compliance in jurisdictions that require algorithmic systems to provide human-readable explanations of consequential decisions [6, 9, 15, 18, 20].

Liquidity and scalability. Results are reported for a portfolio of moderate size ($\sim \$10\text{M}$ notional). Market impact will increase non-linearly with capital, particularly for the cryptocurrency and small-cap equity components of the universe, and the reported Sharpe ratios may degrade at institutional scale.

5.3 Comparison with Large Language Model-Based Approaches

Recent work on LLM-based trading agents [17, 19] has demonstrated that general-purpose language models can synthesize analyst reports, macroeconomic narratives, and earnings transcripts into directional forecasts with competitive accuracy. The proposed system’s FinBERT-based sentiment module is more lightweight but domain-specialized, offering lower inference latency (critical for intra-day signal generation) at the cost of reduced coverage of long-horizon narrative signals. Hybrid architectures that use LLMs for weekly thematic signal generation and FinBERT for intra-day event extraction represent a promising direction for future integration.

6 Conclusion and Future Work

This paper has presented a comprehensive, modular AI-driven trading architecture that integrates pattern recognition, NLP-based sentiment analysis, and deep reinforcement learning within a unified ensemble decision framework. The system incorporates online regime detection, hierarchical risk management, and a latency-aware execution engine, yielding robust performance across equities, ETFs, derivatives, and cryptocurrency markets over a 24-month out-of-sample evaluation period. The multi-factor approach achieves a Sharpe ratio of 2.14, a maximum drawdown of 8.3%, and a cumulative return of 62.4%, substantially outperforming both single-factor momentum and buy-and-hold baselines.

Several directions for future research are identified:

1. **Transformer-based RL.** Replacing the MLP policy network with a transformer-based architecture (e.g., Decision Transformer [19]) could improve the agent’s ability to condition on long-horizon return histories and exploit non-Markovian dependencies.
2. **Explainable AI for compliance.** Integrating SHAP-based feature attribution and counterfactual explanation generation would improve the interpretability of individual trading decisions, facilitating regulatory approval in EU and US markets.
3. **LLM-FinBERT hybrid sentiment.** Combining a large language model for weekly narrative signal extraction with FinBERT for intra-day event detection could enhance sentiment signal coverage without sacrificing latency.
4. **Multi-agent market simulation.** Training the DRL agent against a population of adversarial agents in a simulated order-book environment would improve robustness to strategic counterparties and reduce susceptibility to adversarial manipulation [3, 8, 16, 19].
5. **Live deployment and A/B testing.** A paper-trading pilot followed by a controlled live deployment with randomized strategy allocation (A/B test) would provide definitive evidence of the system’s ability to generate alpha net of all real-world frictions.

References

- [1] K. Pankwaen et al., “Global Cross-Market Trading Optimization Using Iterative Combined Algorithm: A Multi-Asset Approach with Stocks and Cryptocurrencies,” *Mathematics*, 2025.
- [2] H. Goyal et al., “AI-Powered Autonomous Trading: Enhancing Market Efficiency Through Predictive Analytics,” *International Journal for Research in Applied Science and Engineering Technology*, 2024.
- [3] L.-C. Cheng et al., “Multiagent-based deep reinforcement learning framework for multi-asset adaptive trading and portfolio management,” *Neurocomputing*, 2024.
- [4] R. Dey et al., “Machine Learning-Based Automated Trading Strategies for the Indian Stock Market,” *Journal of Information Systems Engineering and Management*, 2025.
- [5] A. Bárdi et al., “Optimizing Algorithmic Trading by Integrating Sentiment Analysis,” in *Proc. 2024 IEEE 11th Int. Conf. on Computational Cybernetics and Cyber-Medical Systems (ICCC)*, 2024.
- [6] R. Leng, “AI-Driven Optimization of Financial Quantitative Trading Algorithms and Enhancement of Market Forecasting Capabilities,” *Applied and Computational Engineering*, 2024.
- [7] G. Cohen, “Algorithmic Trading and Financial Forecasting Using Advanced Artificial Intelligence Methodologies,” *Mathematics*, 2022.
- [8] L. Avramelou et al., “Deep reinforcement learning for financial trading using multi-modal features,” *Expert Systems with Applications*, 2023.
- [9] V. Pillai, “Integrating AI-Driven Techniques in Big Data Analytics: Enhancing Decision-Making in Financial Markets,” *International Journal of Engineering and Computer Science*, 2023.
- [10] L. Cagliero et al., “Shortlisting machine learning-based stock trading recommendations using candlestick pattern recognition,” *Expert Systems with Applications*, 2023.
- [11] W.-C. Chiang et al., “An adaptive stock index trading decision support system,” *Expert Systems with Applications*, 2016.
- [12] T.-L. Chen et al., “An intelligent pattern recognition model for supporting investment decisions in stock market,” *Information Sciences*, 2016.
- [13] B. Dong et al., “Real-time Early Warning of Trading Behavior Anomalies in Financial Markets: An AI-driven Approach,” *Journal of Economic Theory and Business Management*, 2025.
- [14] R. R. Patil, “AI-Infused Algorithmic Trading: Genetic Algorithms and Machine Learning in High-Frequency Trading,” *International Journal For Multidisciplinary Research*, 2023.
- [15] M. Dokumacı, “AI in Forecasting Financial Markets,” *Human Computer Interaction*, 2024.
- [16] M. Patil et al., “AI driven dynamic Prediction System for Stock Market Trends using Multimodal Data and Adaptive Machine Learning Models,” *International Journal of Research Publication and Reviews*, 2025.
- [17] M. Wawer et al., “Integrating Traditional Technical Analysis with AI: A Multi-Agent LLM-Based Approach to Stock Market Forecasting,” 2025.

- [18] W. A. Addy et al., “Algorithmic Trading and AI: A Review of Strategies and Market Impact,” *World Journal of Advanced Engineering Technology and Sciences*, 2024.
- [19] C. Zhang et al., “When AI Meets Finance (StockAgent): Large Language Model-based Stock Trading in Simulated Real-world Environments,” *arXiv*, 2024.
- [20] T. Sahani, “Decoding Market Emotions: The Synergy of Sentiment Analysis and AI in Stock Market Predictions,” *Journal of Next-Generation Research 5.0*, 2024.